Sahaj Bhadja

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Summary

Software Engineer with 2+ years building high-performance user applications with **Kotlin/Android-adjacent**, **React/TypeScript**, and backend APIs, focused on clean architecture, debugging, and scalable mobile-style experiences.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed and shipped high-throughput user workflows in a **Flask + React** application handling **10K+ daily requests**, with modular APIs and performance-focused UI behavior.
- Improved concurrent processing and release reliability via **GitHub Actions CI/CD**, reducing turnaround time by **60%** while maintaining stable runtime under peak load.

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built backend services for production user applications across **Azure SQL/PostgreSQL**, reducing dashboard latency from **1.2s to 350ms** through schema and query optimization.
- Implemented robust validation/release pipelines for **5K+ monthly jobs** with reconciliation and defect prevention checks to improve reliability.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered API and ETL workflows with **OOP**, AWS Glue, and S3, improving response times by **30%** and pipeline efficiency by **68%**.
- Maintained **20+ unit/integration suites** (Jenkins, PyTest, Mockito) across 3 OS environments, strengthening regression confidence and release quality.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built reproducible simulation pipelines in **Python** processing **100K+ trajectory logs**, with structured debugging and validation to support rapid iteration.
- Documented model behavior and issue patterns for cross-team reviews, improving clarity in technical discussions and risk communication.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built user-facing modules with **REST APIs** and **WebSockets** supporting 8+ workflows and 2M+ synchronized records with responsive, stable behavior.
- Automated deployment/testing through **Jenkins + Docker**, reducing deployment time by **50%** and improving release consistency.

Skills

Languages: **Kotlin familiarity**, Java, Python, JavaScript/TypeScript, SQL, C++
Mobile & App Dev: Android development concepts, immutable-data/functional-pattern awareness, responsive UI architecture, **Jetpack Compose familiarity**
Frontend/Backend: React.js, HTML/CSS, Flask, FastAPI, Node.js, REST APIs, JSON
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL
Tooling: Git, GitHub Actions, Jenkins, Docker, Kubernetes, CI/CD, code reviews
Engineering: Performance optimization, debugging, test automation, agile/scrum collaboration, technical communication

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio*
- Built high-usage healthcare workflows with structured API integration and UX-driven flows supporting **100+ daily appointments** and **2x faster** record retrieval.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built secure, high-performance user workflows for **50K+ users** and **1M+ transactions** with real-time updates and scalable architecture.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **6 scheduling algorithms** with interactive controls and live visualization across **1K+ process combinations**, strengthening performance reasoning and systems debugging.